

In[14]:= **MatrixForm**[A = {{1, 2}, {3, 4}, {5, 6}}]

Out[14]//MatrixForm=

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$$

In[6]:= **xr** = **Table**[{**Random**[**Real**, {0, 1}], **Random**[**Real**, {0, 1}]}], {10}]

Out[6]= {{0.923634, 0.0820782}, {0.633435, 0.505834},
{0.658398, 0.800711}, {0.408144, 0.437489}, {0.046357, 0.0919652},
{0.537779, 0.677918}, {0.0335325, 0.655447},
{0.245078, 0.159239}, {0.732237, 0.057886}, {0.560766, 0.890746}}

In[13]:= **yr** = **Table**[A.xr[[i]], {i, **Length**[xr]}]

Out[13]=

{{1.08779, 3.09921, 5.11064}, {1.6451, 3.92364, 6.20218},
{2.25982, 5.17804, 8.09626}, {1.28312, 2.97439, 4.66565},
{0.230287, 0.506932, 0.783576}, {1.89362, 4.32501, 6.7564},
{1.34443, 2.72238, 4.10034}, {0.563556, 1.37219, 2.18082},
{0.848009, 2.42826, 4.0085}, {2.34226, 5.24528, 8.1483}}

In[17]:= **pinv** = **PseudoInverse**[xr]

Out[17]=

{{0.564187, 0.170463, 0.0447928, 0.0561169, -0.0139823,
0.0250799, -0.293687, 0.0835135, 0.450734, -0.0623743},
{-0.386467, 0.0541209, 0.251455, 0.114078, 0.0429711,
0.222355, 0.449201, -0.0049531, -0.311483, 0.362388}}

In[18]:= **Dimensions**[pinv]

Out[18]=

{2, 10}

In[21]:= **Dimensions**[yr]

Out[21]=

{10, 3}

In[20]:= **Dimensions**[A]

Out[20]=

{3, 2}

In[29]:= **MatrixForm**[AA = **Transpose**[yr].**Transpose**[pinv]]

Out[29]//MatrixForm=

$$\begin{pmatrix} 1. & 2. \\ 3. & 4. \\ 5. & 6. \end{pmatrix}$$